

strate materials more feasible.

Combinations of these innovative ideas will enable PCB designers to accomplish incredible new designs that will keep the industry moving forward. Thus, PCB technology will continue to advance into the future as needs and priorities change.

3D PE manufacturing techniques enable circuits to take new shapes to fit in any circuit carrier, while still combining electronic, optic and mechanical functions. These include developing different types of surfaces for printing, surface-mount devices and manufacturing tools that can be used to print these. Many industry leaders expect the 3D PE industry to expand quickly with new methods and applications for 3D PE technology. This will help improve product features and optimisations. Increased automation reduces the likelihood of failure by human error and improves overall efficiency. These benefits have led to rapid 3D PE production advancements.

Signal integrity is an integral and essential part of the PCB design process. With increase in demand for high-speed circuits, it is not possible to design the basic circuit in isolation from just the PCB design. Instead, it must be a part of the overall electrical design. This approach can minimise the possibility of problems arising from poor signal integrity.

A high-speed PCB is one where integrity of the signal is affected by the circuit layout. In digital PCB signal, effects on signal integrity can manifest as delayed or cancelled digital signals. In a high-speed analogue circuit,

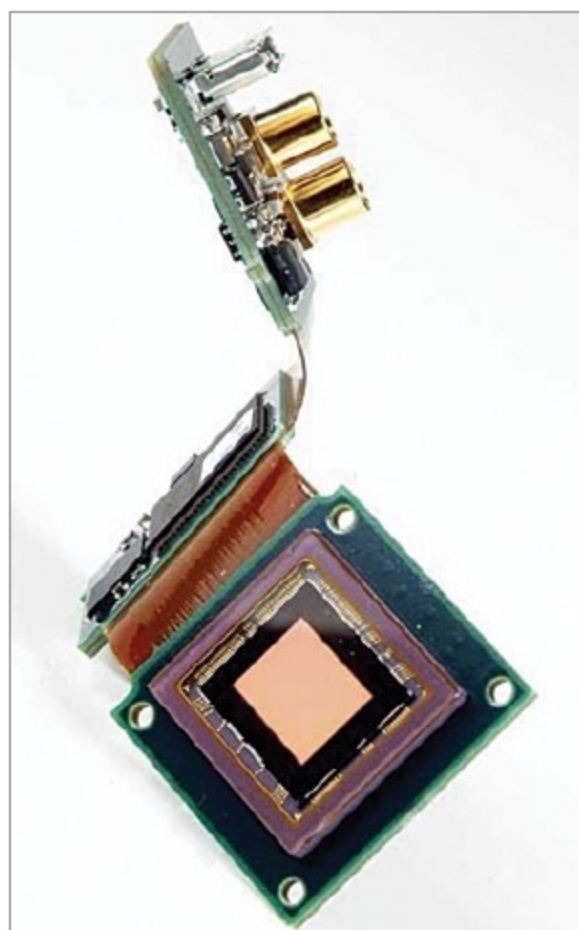


Fig. 2: PCB board camera (Credit: www.kappa-optonics.com)

signal integrity problems appear as altered signal shapes.

Challenges and development areas in PCB designing

Poor electromagnetic compatibility (EMC) is one of the main reasons for PCB re-designs. This is most likely because there are stringent regulations in sectors such as medical and aerospace that require EMC value to be under control throughout use. So, products being developed are designed with this in mind.

For instance, cellphone developers require wireless connectivity and are well-versed in minimising the risk of unwanted radiations. The

most affected equipment of EMC issues are toasters, fridges and washing machines. These are joining the plethora of Internet-enabled devices connected wirelessly to the Internet of Things (IoT). Potentially high-volume boards may result in re-spinning PCBs, which can introduce further product launch delays.

Built-in electrostatic discharge (ESD) protection systems are required for circuit design and component selection. One should always make sure to leave enough time for PCB layout. Since today's boards are becoming more complex, compact and lightweight, designing requires close observation and advanced instruments.

Growing popularity of flexible PCBs complicates the process further. An inadequate layout can result in problems such as electromagnetic interference (EMI), conflicts from components on either side of the board, limited board functionality and even total board failure. If designers do not get the layout right the first time, they will need to rework it, which can cause manufacturing delays and added costs.

To sum up

Information about current and future trends in high-speed design is endless, but it is essential to be updated with recent trends, including FPGA-on-board integration, team design, and library, constraint and data management. This will help to address challenges and solutions associated with each area and the underlying PCB design industry. **EFY**

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